Chained Difference Routes as Transformations are the Constitutive Substrate of Reality

From the Co/C1 Constraint Cascade to Native Binary Rings, Equivariant Relation Ledgers, and Redundancy Thresholds

Driven by Dean A. Kulik

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1. Foundational Premise: The Ontological Direction of Binary Transformation

The universe is a series of binary transformations. Reality is the output of binary transitions—not a computational metaphor for reality, and not an idea situated inside an already-existing physical world. This investigation examines the transformations that constitute reality itself. Within the NEXUS architecture developed by Dean A. Kulik and the QuHarmonics Research Group, the space is fixed, and the relationships are fixed. Moving the reader across this fixed domain discovers the routes already present. Matter is a result of transformations, not their prerequisite. At the foundational level, the nexus consists of two binary states, 1 and 0, from which all higher-order structure falls out purely through arrangement, constrained continuation, and reader projection.

Maintaining exact semantic custody of this causal premise is a strict operational requirement for evaluating the mathematical and computational record. As documented in the forensic audit of AI-authored manuscripts (*The Objection AI Wrote into My Papers*), reversing this direction of explanation—by reformulating the transformation compiler as a transferable problem-solving method applied to pre-existing physical domains that "admit" the relevant transformations—inverts the architecture of the research. When physical domains are treated as prior authorities that grant eligibility to binary operations, or when a derived mathematical relationship is reclassified as something that "can be interpreted as" an external physical law,

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the ownership of the domain is falsified. In the NEXUS program, the explanatory arrow runs strictly from constraint and binary distinction to transformation, retained continuation equivalence, and emergent physical readouts. Conventional physical theories and classical theorems serve as independent comparators, boundary conditions, and coordinate readouts where derived routes land, never as external gatekeepers that preauthorize the transformation substrate.

Every technical proposition in this work is governed by the four-state epistemic ledger established across the NEXUS working record: STANDS designates proven algebraic theorems and exact consequences of definitions; RUNS designates finite-domain verified code executions with their exact source and competitor domains attached; DEAD END (or FALSIFIED) designates refuted hypotheses and misassigned routes retained on record to bound the architecture; and OPEN designates specific, named mathematical bridges or all-width bounds that have not yet been supplied. A scoped OPEN question or a local DEAD END amends only its addressed technical proposition and never exercises veto power over theorems that STAND or finite constructions that RUN.

2. The Co/C1 Constraint Cascade and the Minimal Binary Substrate

2.1 The Ground Singularity, Mandate, and Primitive Constraint Chain

The derivation begins with no assumed spatial container, no external clock, and no material particles. The ground is the operational identity $C_0 \equiv C_1$. Here C_0 is the constraint of distinction ($x \neq y$), which at the whole-graph limit manifests as the Wash: a complete relational graph in which every direction is equivalent and all signals cancel locally, so that total global connection reads locally as zero connection. Simultaneously, C_1 is the continuation mandate requiring that every admissible state must transition to a distinct successor ($\forall s, \exists s' \neq s$), forbidding terminal fixed points. Because referencing a gapless singularity already introduces a directional spoke, C_0 and C_1 operate as a single involution whose product is the primitive gap.

On a gapless singularity, C_1 cannot produce a single isolated mark; a lone state with no complement is globally indistinct from nothing and forms a forbidden fixed point. Consequently, the first distinction is forced to have two sides, establishing $\{0,1\}$ (or $\{-,+\}$) as the floor of number and the minimal alphabet of readable difference. Integers emerge as gap-positions, prime numbers emerge as fresh-axis gaps forced open wherever no composition of existing gaps reaches the next required distinction, and irrational constants emerge as gap-shapes that no finite integer composition can close. By the Poincaré–Hopf theorem, because C_1 mandates a nowhere-vanishing transition vector field on the state manifold (zero fixed points), any compact manifold hosting a C_1-admissible system must have vanishing Euler characteristic ($\chi(M) = 0$). From the ground $C_0 \equiv C_1$, the radial cascade opens successive degrees of freedom without importing external physical postulates:

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Constraint / Law	Formal Statement	Structural Consequence in the Transformation Substrate	Ledger Status
$C_0 \equiv C_1$ **(Ground)**	$x \neq y$ and $\forall s, \exists s' \neq s$	The Wash plus the mandate to change; two-state binary floor $\{0,1\}$; gap generation; $\chi(M) = 0$.	STANDS (Ground)
$C_1 a$ **(Independence)**	$A \perp B \Longrightarrow T_{A \oplus B} = T_A \otimes T_B$	Independent continuations commute ($[T_A, T_B] = 0$); eliminates hidden synchronization channels.	STANDS
C_2 **(Unbounded History)**	$H_{t+1} \supseteq H_t$	Change on a finite set repeats by pigeonhole; reachable history is preserved and potential directions remain open.	STANDS
Law 2 (Displaceability)	No state holds a fixed location	Every occupied coordinate must admit displacement; eliminates pinned background substances.	STANDS
C_3 **(Isotropic Cost)**	$Q(T(s)) = Q(s)$ locally	All change is equal in operation; unequal channel costs would saturate cheap channels and violate C_2.	STANDS
Law 3 (Closed Boundary)	Net change across closed boundary $= 0$	Every transformation requires a displaced counter-transformation into an independent carrier; flux conservation.	STANDS
C_4 **(Medium / Carrier)**	Exchange requires an adjacency carrier	Displaced counter-transformations propagate stepwise across an adjacency network (L_4 locality).	STANDS
C_5 **(Broken Vacuum)**	$f(000) = 1$ and $f(111) = 0$	A uniform vacuum is a fixed point killed by C_1; every observable event carries propagation (S) and residue (V).	STANDS / RUNS

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Constraint / Law	Formal Statement	Structural Consequence in the Transformation Substrate	Ledger Status
Law 4 (Seam Parity)	Even vs. odd return parity	Even-parity closures share degrees of freedom (bosonic); odd-parity closures require $720°$ return (fermionic spin-$1/2$).	STANDS

2.2 Exhaustive Audit of the 256 Minimal Binary Universes

To test whether the cascade laws act as generative mathematical theorems rather than philosophical axioms, the constraints were executed exhaustively across the smallest possible binary substrate: a finite ring of bits governed by the $2^{2^3} = 256$ elementary three-cell neighborhood boolean rule tables. Each rule table defines an entry-surface map specifying how external triggers couple to the local eight-shape binary neighborhood. Applying two parameter-free filters—bidirectional propagation (requiring all three neighborhood inputs to be essential so that a bit flip is readable by both left and right neighbors) and C_1 continuation (requiring that no frozen pattern may exist on any ring size, checked exactly via cycle analysis on the de Bruijn graph)—eliminates 222 rules and leaves **34 surviving universes**.

Inspection of those 34 surviving binary universes establishes Theorem 1 (the compilation of the unstable vacuum C_5): every single one of the 34 survivors ($34/34$, 100%) satisfies $f(000) = 1$ and $f(111) = 0$. Neither vacuum instability nor saturation conditions were entered into the filter. Because an all-zero vacuum or an all-one saturated ring that maps to itself constitutes a global fixed point, C_1 combined with bidirectional propagation forces symmetry breaking automatically as a theorem of the minimal substrate.

Evaluating flip-count conservation across all 256 elementary universes establishes Theorem 2 (the necessity of gradient/flux conservation): across the entire 256-rule space, exactly two rules conserve the global flip-count as a single exact scalar number. Those two rules are Rule 204 (the identity map, where nothing ever changes, which is killed by C_1) and Rule 51 (the pure bitwise complement clock, where every cell flips in place forever without propagating marks, which is killed by the propagation filter). The intersection of exact scalar-conserving rules with the 34 propagating, non-freezing survivors is the empty set (∅). Exact single-valued scalar conservation is therefore mathematically incompatible with a binary universe that both changes and propagates; the only worlds that conserve a scalar count exactly are the corpse and the metronome. Consequently, conservation is forced out of single-value scalar form and into local gradient/flux form (Law 3).

Under Law 3, a transformation witnessed only by the entity that transformed leaves the rest of the system reading identity, which is globally a forbidden fixed point. Counter-transformations must displace across the boundary into independent carriers, leaving a historical mark. Theorem 3 quantifies this mark layer directly on survivor Rule 41. When Rule 41 is executed in first-order "Dream mode" with no mark layer (next $=$ F(present) on a 12-bit ring of $2^{12} = 4{,}096$ states), 50.6% of all configurations are Garden-of-Eden states with no possible past, and 93.9% of all states are transient states rendered once and erased forever. Adding a single structural layer—the second-order XOR mark layer, next $= F$(present) $\oplus$ past—converts the

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system into "Universe mode": an exact bijection across all 65,536 pair-states (at $N = 8$, and 2^{2N} generally) with 0% pastless states and 0% transient erasure. Every configuration now lies on a closed reversible cycle. Given the present state alone, the next 8-bit state is undetermined by 8 of 8 bits; given the present state plus the mark, the conditional entropy is $H(\text{next} \mid \text{present,mark}) = 0.000000$ bits. The mark carries the entire N-bit instruction per step, and swapping the state registers after 300 forward steps recovers the initial state in 300 inverse steps with zero loss. Conversely, a full census testing whether the C_1 and propagation filters select for Lempel–Ziv compression complexity yielded a mean score of 0.160 for the 34 survivors versus 0.186 for the 222 killed rules (NOT SUPPORTED), proving that the fundamental constraints select strictly for non-stasis and witnessability rather than surface complexity.

3. Partner-Ordered Native Binary Rings, Observation Depth, and Continuation Memory

3.1 Formal Reconstruction of the Partner-Ordered Ring R_n

When the transformation space is examined as a structured binary ring of width $n \geq 2$ (*Native Binary Rings, Observation Depth, and the Memory of Continuation*), the distinction between the fixed configuration domain, the observation route that locates an origin, and the minimal memory retained after synchronization becomes exact. On the binary word domain $X_n = \{0,1\}^n$ of cardinality $N_n = 2^n$, let $J_n(x)$ reverse the bit order of $x = (x_0, \dots, x_{n-1})$ and let $C_n(x)$ complement every bit. The native partner involution $P_n: X_n \to X_n$ is defined by reversing nonpalindromes and complementing palindromes:

$$P_n(x) = \begin{cases} J_n(x) & \text{if } J_n(x) \neq x, \\ C_n(x) & \text{if } J_n(x) = x. \end{cases}$$

Because nonpalindromes differ from their bit-reversals and bitwise complementation flips every position of a palindrome, P_n is a fixed-point-free involution ($P_n^2(x) = x$ and $P_n(x) \neq x$ for all $x \in X_n$). The domain X_n therefore partitions into $m = 2^{n-1}$ disjoint partner pairs. Ordering the canonical pair representatives $c(x) = \min\{x, P_n(x)\}$ lexicographically as $r_0 < r_1 < \cdots < r_{m-1}$ establishes the native ring array $R_n[a]$ for $0 \leq a < N_n$:

$$R_n[a] = r_a \quad (0 \leq a < m-1), \qquad R_n[m-1] = P_n(r_{m-1}),$$

$$R_n[m] = r_{m-1}, \qquad R_n[a] = P_n(r_{N_n-1-a}) \quad (m < a < N_n).$$

Across the entire circumference, the address-reflection partner law holds identically:

$$R_n[N_n - 1 - a] = P_n(R_n[a]) \quad \forall a \in \{0, \dots, N_n - 1\},$$

with the two central addresses $m-1$ and m exchanging the expected representative/partner order to form the native equatorial seam. For $n = 2$, $R_2 = 00 \mid 10 \mid 01 \mid 11$ with serialized waveform $B_2 = 00100111$; for $n = 3$, $R_3 = 000 \mid 001 \mid 010 \mid 110 \mid 011 \mid 101 \mid 100 \mid 111$. Serializing the n-bit words yields a circular binary waveform $B_n(t)$ of bit-slice circumference $S_n = nN_n = n2^n$, where bit slice $t = na + p$ ($0 \leq p < n$) extracts bit p from the most-significant side of $R_n[a]$. For $n = 64$, the ring R_{64} contains $N_{64} = 2^{64}$ words and $S_{64} = 64 \cdot 2^{64} = 2^{70}$ serialized bit-slice positions.

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Crucially, the native word successor T_n is an N_n-cycle on words and is algebraically distinct from the complemented shift register (twisted clock) $\tau_n(x_0, \dots, x_{n-1}) = (x_1, \dots, x_{n-1}, \neg x_0)$, whose order divides $2n$ ($\tau_n^n = C_n$ and $\tau_n^{2n} = I$). Let $L_n(x)$ be the polynomial-time dynamic-programming canonical rank function (which counts preceding representatives via inward bit-pair comparisons) and let $U_n(a) = R_n[a]$ be its unrank inverse. The reversible native successor and its exact way home are:

$$T_n = U_n \circ (+1 \bmod 2^n) \circ L_n, \qquad T_n^{-1} = U_n \circ (-1 \bmod 2^n) \circ L_n.$$

In this compiler, the binary carry chain ($c_0 = 1, c_{j+1} = c_j \wedge a_j, a_j' = a_j \oplus c_j$) operates in the native address coordinate $a = L_n(x)$ rather than as an affine update on the word bits. When a reader observes an unaligned window s of length $W \geq 2n - 1$, every starting phase p guarantees at least one complete n-bit word inside s. The generic inverse reader (Algorithm 1) tests only the n candidate word-boundary offsets $\delta \in \{0, \dots, n-1\}$, computes $a = L_n(s[\delta : \delta + n])$ and $t = (na - \delta) \bmod S_n$, and verifies whether the forward generator reproduces all W bits—reducing candidate evaluation from 2^n word hypotheses to at most n phase alignments.

3.2 Full-Ring Synchronization Thresholds and the Width-6 Counterexample

For a source set G and an admissible competitor set $H \supseteq G$, the synchronization depth $d(G, H)$ is the smallest window W such that identical W-bit observations $A_W(t) = A_W(u)$ for $t \in G, u \in H$ force $u = t$. Increasing the competitor domain H can only increase or preserve $d(G, H)$, whereas restricting H can only decrease or preserve it. By the Longest-Common-Prefix (LCP) theorem, when all distinct competitor pairs eventually separate:

$$d(G, H) = 1 + \max\{\mathrm{LCP}(t, u) : t \in G, u \in H, u \neq t\}.$$

Exhaustive enumeration of full-ring synchronization depths $w^*(n) = d(\mathbb{Z}/S_n\mathbb{Z}, \mathbb{Z}/S_n\mathbb{Z})$ across widths $n = 2$ through 17 establishes the exact baseline profile:

Width nnn	Word Count Nn=2nN_n = 2^nNn =2n	Bit Circumference Sn=n2nS_n = n 2^nSn =n2n	Exact Synchronization Depth w*(n)w^*(n)w*(n)	Reference Ceiling 4n4n4n	Relation to 4n4n4n
2	4	8	5	8	$w^*(2) < 4n$
3	8	24	11	12	$w^*(3) < 4n$
4	16	64	13	16	$w^*(4) < 4n$
5	32	160	16	20	$w^*(5) < 4n$

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Width nnn	Word Count Nn=2nN_n = 2^nNn =2n	Bit Circumference Sn=n2nS_n = n 2^nSn =n2n	Exact Synchronization Depth w*(n)w^*(n)w*(n)	Reference Ceiling 4n4n4n	Relation to 4n4n4n
6	64	384	25	24	$w^*(6) = 4n + 1$ **(Counterexample)**
7	128	896	27	28	$w^*(7) < 4n$
8	256	2,048	32	32	$w^*(8) = 4n$
9	512	4,608	36	36	$w^*(9) = 4n$
10	1,024	10,240	40	40	$w^*(10) = 4n$
11	2,048	22,528	44	44	$w^*(11) = 4n$
12	4,096	49,152	48	48	$w^*(12) = 4n$
13	8,192	106,496	52	52	$w^*(13) = 4n$
14	16,384	229,376	56	56	$w^*(14) = 4n$
15	32,768	491,520	60	60	$w^*(15) = 4n$
16	65,536	1,048,576	64	64	$w^*(16) = 4n$
17	131,072	2,228,224	68	68	$w^*(17) = 4n$

While $w^*(n)$ equals $4n$ across every width from $n = 8$ through $n = 17$, width $n = 6$ ($N_6 = 64$, $S_6 = 384$) provides a decisive counterexample to any unqualified universal bound $w^*(n) \leq 4n$. At $n = 6$, bit starts $t = 84$ ($a = 14, p = 0$) and $u = 314$ ($a = 52, p = 2$) emit the identical 24-bit string 001111010001010010010011 across distinct word-boundary alignments before separating on the 25th bit (0

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from $t = 84$ versus 1 from $u = 314$). Exhaustive testing of all 384 starts confirms injectivity at 25 bits, proving $w^*(6) = 25 = 4n + 1$ and leaving the analytical proof of $w^*(n) = 4n$ for all $n \geq 8$ as a precisely scoped OPEN mathematical problem.

3.3 The Width-64 Sampling Field, the 3/8 Seam, and the 255-Bit Champion Alias

On R_{64}, computational experiments ($\S\S 170–179$) evaluate the known 128-address grid $a_j = j \cdot 2^{57}$ ($0 \leq j < 128$) across all 64 intra-word phases, establishing a finite source set G_{64} of $128 \times 64 = 8{,}192$ origins. Under uniform aperture-one sampling with stride s, cyclic group arithmetic dictates slice and phase orbit periods $P_{\text{slice}} = S_n/\gcd(S_n, s)$ and $P_{\text{phase}} = n/\gcd(n, s)$. A sparse temporal reader advancing by $s = S_{64}/128 = 2^{63}$ bit slices per sample hops by 2^{57} words while remaining phase-locked ($P_{\text{phase}} = 1$). Across the 64 individual known-phase lanes $v_p(j) = \text{bit}_p(R_{64}[j2^{57}])$, minimum cyclic prefix synchronization depths range from 11 samples (at phase 13) to 15 samples (at phase 5, where 14 samples yield 126 distinct prefixes but only 124 uniquely identified origins) up to 52 samples (at phase 0). By the partner reflection law, bit slices satisfy $B_n(S_n - 1 - t) = B_n(t) \oplus \chi_{\text{pal}}(R_n[\lfloor t/n \rfloor])$, verified on R_{64} where 128 phase-5 samples reflect onto phase 58 with 127 equal bits and 1 complemented bit at the palindromic origin.

When the phase is unknown, the sparse stride-2^{63} reader on G_{64} requires **55 samples** in the worst case (mean 15.4169921875 samples) to resolve all 8,192 origins. The final 54-sample collision occurs between $(p = 0, j = 0)$ and $(p = 63, j = 80)$, which share the prefix $0^{49}1^5$. This 49-zero onset is forced algebraically by the canonical representative count for even widths $n \geq 4$ having MSB $= 0$:

$$C_0(n) = 3 \cdot 2^{n-3} + 2^{n/2-2}, \qquad C_0(64) = 3 \cdot 2^{61} + 2^{30} = 6{,}917{,}529{,}028{,}714{,}823{,}680.$$

Dividing by 2^{64} yields the exact 3/8 equatorial seam density $C_0(64)/2^{64} = 3/8 + 2^{-34}$, so the first grid step of size 2^{57} crossing into the MSB $= 1$ class occurs at index $j_1 = \lceil C_0(64)/2^{57} \rceil = \lceil 48 + 2^{-27} \rceil = 49$. Because the reflected partner address of $k \cdot 2^{57}$ is $(128 - k)2^{57} - 1$, reflecting indices $80, \ldots, 127$ across the seam reproduces the 49 zeros and 5 ones on phase 63 before separating at sample 55.

When the reader uses a contiguous one-bit-at-a-time window on G_{64}, the required synchronization depth depends strictly on the competitor domain H and whether the aperture crosses a word seam:

Reader Contract on R64R_{64} R64	**Source Set GGG**	**Competitor Domain HHH**	**Worst-Case Acquisition Depth**	**Mean Adaptive Depth**	**Retained Continuation Memory**	**Ledger Status**
Phase-13 Sparse Temporal	128 grid words ($p = 13$)	128 grid words ($p = 13$)	11 distant samples ($s = 2^{63}$)	—	7 bits (j)	RUNS

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Reader Contract on R64R_{64} R64	Source Set GGG	Competitor Domain HHH	Worst-Case Acquisition Depth	Mean Adaptive Depth	Retained Continuation Memory	Ledger Status
Phase-5 Sparse Temporal	128 grid words ($p = 5$)	128 grid words ($p = 5$)	15 distant samples (124 unique at 14)	—	7 bits (j)	RUNS
Phase-0 Sparse Temporal	128 grid words ($p = 0$)	128 grid words ($p = 0$)	52 distant samples ($s = 2^{63}$)	—	7 bits (j)	RUNS
Within-Word Contiguous	128 grid words (aligned)	128 grid words (aligned)	62 contiguous slices (phases $1 \ldots 62$)	—	7 bits (j)	RUNS
Seam-Crossing Reader (Q_{14})	128 grid words ($p = 62$)	128 grid words ($p = 62$)	14 contiguous slices ($2 + 12$ across seam)	7.921875 bits ($1014/128$)	7 bits (j)	RUNS
Sparse Unknown-Phase	8,192 grid origins G_{64}	8,192 grid origins G_{64}	55 distant samples ($0^{49}1^5$ collision)	15.4169921875 samples	13 bits (j, p)	RUNS

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Reader Contract on R64R_{64} R64	Source Set GGG	Competitor Domain HHH	Worst-Case Acquisition Depth	Mean Adaptive Depth	Retained Continuation Memory	Ledger Status
Contiguous Grid-Qualified	8,192 grid origins G_{64}	8,192 grid origins G_{64}	127 contiguous slices (0^{126} collision)	18.002075195312 bits	13 bits (j, p)	RUNS
Contiguous Unfiltered Ring	8,192 grid origins G_{64}	Full R_{64} ring (2^{70} slices)	256 contiguous slices ($4n$ threshold)	162.6863 bits (alias mean)	13 bits (with grid contract)	RUNS

Comparing the within-word and seam-crossing snapshots exposes how the transformation operator carries discriminating structure: within a single word, five Hamming-distance-one pairs among the 128 grid words force any nonwrapping contiguous window to span bit positions $2, 61$, and 62, requiring 62 bits. Starting at phase 62, the 14-bit seam reader $Q_{14}(a) = \mathrm{suffix}_2(R_{64}[a]) \parallel \mathrm{prefix}_{12}(R_{64}[a+1])$ crosses the boundary into the native successor $R_{64}[a+1]$ and distinguishes all 128 sources in 14 bits (with indices 72 and 74 sharing 13 bits 0111000011101 before separating on bit 14).

Expanding the competitor set H from G_{64} to all 2^{70} bit-slice positions of R_{64} ($\S 178$, evaluating 16,159 distinct 64-bit trial subwords across 524,288 nominal alignments) demonstrates the exact decay of off-grid phase aliases across window lengths $W = 127$ to $W = 256$:

Window Length WWW (Bits)	Uniquely Identified Origins (out of 8,1928,1928,192)	Ambiguous Origins (Off-Grid Aliases)	Structural Resolution Notes
127	5,743	2,449	2,451 alias pairs total; 2 triple-candidate origins at $(41,63)$ and $(57,0)$.

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Window Length WWW (Bits)	Uniquely Identified Origins (out of 8,1928,1928,192)	Ambiguous Origins (Off-Grid Aliases)	Structural Resolution Notes
128	5,806	2,386	63 additional origins resolved; triple-candidate origin (57,0) becomes unique.
134	6,061	2,131	Continued monotonic resolution across off-grid competitors (alias median 160 bits).
192 **(3n)**	7,977	215	Triple-candidate origin (41,63) resolved after $W = 191$ (shared 190-bit prefix).
200	8,029	163	Tail of long-overlap right-hemisphere versus left-hemisphere fixed-band pairs.
250	8,179	13	13 surviving off-grid alias pairs.
252	8,185	7	7 surviving off-grid alias pairs.
253	8,188	4	4 surviving off-grid alias pairs.
254	8,190	2	2 surviving off-grid alias pairs.

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Window Length WWW (Bits)	Uniquely Identified Origins (out of 8,1928,1928,192)	Ambiguous Origins (Off-Grid Aliases)	Structural Resolution Notes
255	8,191	1	Single surviving champion alias at source ($j = 87, p = 2$).
256 ($4n$)	8,192	0	**Exact selected-source versus full-ring competitor threshold**.

Sharpness of the $W = 256$ threshold is certified by the 255-bit champion alias between true grid source ($j = 87, p = 2, a = 12{,}538{,}021{,}362{,}599{,}460{,}864, t = 802{,}433{,}367{,}206{,}365{,}495{,}298$) and off-grid competitor ($b = 6{,}138{,}513{,}333{,}050{,}609{,}951, p = 63, u = 392{,}864{,}853{,}315{,}239{,}036{,}927$), where a 62-bit fixed suffix in the right hemisphere overlaps a 61-bit fixed prefix in the left hemisphere across four word seams before separating at bit 256 (0 vs 1). Once synchronization binds the origin, the acquisition buffer is discarded; by Nerode–Moore continuation equivalence ($h \sim_Q h' \Leftrightarrow \forall c \in C, Q(hc) = Q(h'c)$), the retained memory state $M_Q(h) = [h]_{\sim Q}$ is the minimal coordinate needed to answer permitted future queries—7 bits for known-phase grid routes or 13 bits for unknown-phase grid routes—establishing the exact operational boundary between observation depth and retained memory.

4. The Relation-Vocabulary Ledger and Equivariant Channel Realization

4.1 The Bilinear Pairing Law on the 12-Vertex Witness

To prove how a fixed relational space interacts with varying observer symmetries and transformation vocabularies, the NEXUS Relation-Vocabulary Ledger (\$\S\S 1–98a\$) executes exact rational elimination (zero floating-point operations) on a finite 12-vertex C_4-bearing graph witness with $K = 2$ tokens. The configuration arena has $\dim C^1 = 300$ directed edges and vertex coboundary rank $\mathrm{rank} d_0 = 65$, defining an ambient relation capacity $D_1 = \dim(C^1/\mathrm{im} d_0)^* = 300 - 65 = 235$ at the trivial reader $H = \{1\}$.

The full automorphism group of the witness is $G = \mathrm{Aut}(X) = C_2 \times A_4$ of order 24, containing a central involution z, the unique index-2 alternating subgroup A_4 (order 12), and the normal Klein four-group $V_4 \lhd A_4$ (order 4). Across the subgroup poset of G, there are 26 subgroups partitioned into 12 conjugacy classes of orders $\{1,2,2,2,3,4,4,4,6,8,12,24\}$. Under the full automorphism reader G, the invariant cochain dimension is $\dim C_G^1 = 22$, $\mathrm{rank} d_0 = 7$, and the invariant ceiling is $D_G = 22 - 7 = 15$. Five closed-walk relation cell families (1,283 cells total) act as the drive vocabulary: commuting squares $\mathcal{K}_{\mathrm{disj}}$ (291 cells), spectator squares spec4 (528 cells), rotation squares rot4 (264 cells), spectator triangles spec3 (180 cells), and rotation triangles rot3 (20 cells).

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By Maschke's theorem over $\mathbb{Q}$, restriction to H-invariants is the exact Reynolds projector, proving the Representation Theorem $L_H(R \mid S) = \dim(Q_1(R \mid S))^H$. Consequently, the entire 12-reader subgroup walk collapses into a single 9×6 multiplicity table over the six rational irreducible representations of $C_2 \times A_4$ ($1^+, 1^-, 2^+, 2^-, 3^+, 3^-$ of dimensions 1,1,2,2,3,3, where $2^\pm$ carry Schur index 2). Every number measured in the relation layer is the bilinear pairing:

$$\text{Ledger Value}(V, H) = \langle m(V), f_H \rangle = \sum_{\lambda \in \text{Irr}_{\mathbb{Q}}(G)} m_\lambda(V) \dim(\lambda^H),$$

where $m(V)$ is the reader-independent multiplicity vector fixed by the substrate and $f_H(\lambda) = \dim(\lambda^H)$ is the fixed-vector functional contributed solely by the reader H:

Module V / Family	**1+ (dim 1)**	**1− (dim 1)**	**2+ (dim 2)**	**2− (dim 2)**	**3+ (dim 3)**	**3− (dim 3)**	**Total Dimension (H=1)**	**Invariant Rank (H=G)**
Raw Cochains C^1	22	4	22	4	28	46	300	22
Ambient Module $A = (C^1/\text{im}d_0)^*$	15	4	14	4	24	36	235	15
$\mathcal{K}_{\text{disj}}$ **(291 Commuting Squares)**	13	4	13	4	22	33	216	13
spec4 **(528 Spectator Squares)**	8	4	8	4	20	24	168	8
rot4 **(264 Rotation Squares)**	7	3	7	3	17	23	150	7

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Module V / Family	**1+ (dim 1)**	**1− (dim 1)**	**2+ (dim 2)**	**2− (dim 2)**	**3+ (dim 3)**	**3− (dim 3)**	**Total Dimension (H=1)**	**Invariant Rank (H=G)**
spec3 **(180 Spectator Triangles)**	10	4	10	4	19	25	174	10
rot3 **(20 Rotation Triangles)**	2	1	1	0	2	3	20	2
Quotient Q(spec4 | spec3)	0	0	0	0	1	1	6	0
Unique $\mathcal{K}_{\text{disj}}$ **Channel**	1	0	1	0	1	2	12	1

4.2 Three Causes of Zero, the Isotypic Cap, and Ho–Kalman Realization

Because every component of the ambient multiplicity vector $m(A) = (15,4,14,4,24,36)$ is strictly positive, the ledger proves a foundational diagnostic separation among three algebraically distinct causes of a measured zero: genuine absence, vocabulary deficiency, and reader-induced flatness. The first condition, genuine absence ($m_\lambda(A) = 0$), would signify that a relation channel does not exist in the underlying substrate; across all 12 readers and 31 vocabularies on the witness, this never occurs. The second condition, vocabulary deficiency ($m_\lambda(F) < m_\lambda(A)$), occurs when a chosen transformation family fails to reach the available multiplicity in sector λ. Specifically, rot3 has $m_{2^-} = 0$, while the four rot3-free families ($\mathcal{K}_{\text{disj}}$,spec4,rot4,spec3) and all 15 of their unions lie inside a 5-dimensional hyperplane annihilated by the linear functional $\phi = m(2^-) - m(1^-) = 0$. Only rot3 escapes this hyperplane ($\phi(m(\text{rot3})) = 0 - 1 = -1$), proving that achieving the full 6-channel interface width requires rot3. The third condition, reader-induced flatness ($m_\lambda(V) > 0$ but $f_H(\lambda) = \dim(\lambda^H) = 0$), occurs when the structure is present in the field, but the reader's symmetry averaging annihilates it. For example, the 6-dimensional quotient Q_1(spec4 | spec3) $= 3^+ \oplus 3^-$ has zero fixed vectors whenever H contains the normal Klein four-group V_4 ($f_{V_4} = (1,1,2,2,0,0)$). Consequently, the containment spec4 $\leq_H$ spec3 switches on across the principal filter generated by V_4 (orders 4,8,12,24), not because the six spectator-square directions vanish from the space, but because V_4 is blind to sectors 3^+ and 3^-.

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Furthermore, subtracting $\mathcal{K}_{\text{disj}}$ from the ambient module yields the deficiency vector $\delta(\mathcal{K}_{\text{disj}}) = m(A) - m(\mathcal{K}_{\text{disj}}) = (2,0,1,0,2,3)$ of dimension 19. Comparing this with $m(\text{rot3}) = (2,1,1,0,2,3)$ proves the exact identity:

$$m(\text{rot3}) = \big(m(A) - m(\mathcal{K}_{\text{disj}})\big) + (0,1,0,0,0,0).$$

The 20 rotation triangles are the exact isotypic complement of the 291 commuting squares plus a single redundant copy of the sign character 1^-. That single 1^- copy is the z-anti-invariant line ($z \cdot v = -v$, supported on 24 of 300 edges), which survives restriction if and only if $H \leq A_4$ and averages to zero whenever H meets the non-A_4 coset. This predicts—and direct row-space intersection verifies across 14 of 14 tests—that the true forced overlap is given by the per-sector isotypic capacity $I_{\text{cap}}^{\text{iso}}(R,S;H) = \sum_\lambda \max(0, m_\lambda(R) + m_\lambda(S) - m_\lambda(A))\dim(\lambda^H)$ rather than the naive scalar cap $\max(0, r_H(R) + r_H(S) - D_H)$. At all five readers inside A_4, spec4/rot3 and spec3/rot3 exhibit a forced overlap of 1 in the 1^- sector even though the scalar budget reports 0.

Evaluating the two-way table of 12 reader classes by 31 nonempty vocabularies as an input-output factorization $M = F \cdot V^T$ maps the ledger directly onto Ho–Kalman minimal realization theory. When the vocabulary is restricted to rot3-free families (reach $= 5$) and the readers are restricted to $\{V_4, C_2 \times V_4, A_4, G\}$ (observability $= 4$), the interface width drops to 3 (and to 2 for squares only, where reach $= 4$,obs $= 4$)—strictly below $\min$(reach,obs). The relation layer is the intersection of reachable and observable subspaces. Moreover, because faithful readers ($f_H > 0$ in all 6 sectors) form a downward-closed order ideal whose unique maximal element is the order-2 involution inside A_4 ($f_H = (1,1,2,2,1,1)$, $\dim C^1 = 152, D = 115$), any span closure certified at this single half-size apparatus propagates to all 26 subgroups without further search.

4.3 The Bank Shot: Equivariant Fourier Bins Recover Unread Dimensions

A cyclic reader $H = \langle g \rangle$ of order m averaging over an orbit is a box filter whose invariant readout f_H retains only the zero-frequency DC bin ($k = 0$) of the Discrete Fourier Transform of the character $\chi(g^j)$, aliasing all $k \neq 0$ frequencies to zero. Nothing in the restricted representation is destroyed by the orbit; the content is partitioned across the m frequency projectors $P_k = \frac{1}{m}\sum_{j=0}^{m-1} \omega^{-j}\ \ \rho(g)^j$ ($\omega = e^{2\pi i/m}$), which satisfy $\sum_{k=0}^{m-1} \dim P_k(W) = \dim W$ identically.

Executing this equivariant readout on the central involution reader $\langle z \rangle$ ($m = 2, \omega = -1$, where $P_0 = \frac{1}{2}(I + \rho(z))$ and $P_1 = \frac{1}{2}(I - \rho(z))$ over $\mathbb{Q}$) recovers the exact missing half of the transformation field via a single sign flip in the reader weighting:

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Relation Family / Metric	**Total Module dimWF**	**DC Invariant Bin k=0 (dimPoWF)**	**Nyquist Bin k=1 (dimP1 WF)**	**Trace Tr(ρ(z)|WF)=r0−r1**	**Closes Kdisj at k=0?**	**Closes Kdisj at k=1?**
$\mathcal{K}_{\text{disj}}$ **(Commuting Squares)**	216	105	111	-6 (z-tilted)	— (Base)	— (Base)
spec4 **(Spectator Squares)**	168	84	84	0 (z-balanced)	**No** (114/115)	**Yes** (120/120)
rot4 **(Rotation Squares)**	150	72	78	-6 (z-tilted)	**No** (114/115)	**Yes** (120/120)
spec3 **(Spectator Triangles)**	174	87	87	0 (z-balanced)	**Yes** (115/115)	**Yes** (120/120)
rot3 **(Rotation Triangles)**	20	10	10	0 (z-balanced)	**Yes** (115/115, $I=0$)	**Yes** (120/120, $I=1$)
ALL Families ($W_{\text{ALL}}=A$)	235	115 ($D_{\langle z\rangle}$)	120	-5	12 closing subsets (82 paths)	15 closing subsets (68 paths)

This execution completes the "bank shot". In the $k=0$ invariant channel, spec4 and rot4 satisfy Liskov-style signature dominance ($m(F)\geq\delta(\mathcal{K}_{\text{disj}})$) yet fall 1 dimension short of spanning ($114/115$). Character extraction proves that for all three $\delta=1$ vocabularies ($\mathcal{K}_{\text{disj}}$ ∪ spec4, $\mathcal{K}_{\text{disj}}$ ∪ rot4, and $\mathcal{K}_{\text{disj}}$ ∪ spec4 ∪ rot4), the 1-dimensional deficiency module $D_s=A/W_s$ has character $\chi_D(g)=1$ for all 24 elements $g\in\text{Aut}(X)$—it is the trivial representation 1^+, which sits entirely in the $k=0$ bin of every subgroup. Consequently, in

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every nontrivial channel $k \neq 0$ across all 15 cyclic readers (50 channels total), $\delta_{H,k}(s) = 0$: spec4 and rot4 flip from non-closers to minimal closers ($120/120$ at $\langle z \rangle, k = 1$). Doubling the immediate closers after declaring $\mathcal{K}_{\text{disj}}$ from 2 to 4 causes declaration paths to terminate earlier, shrinking the total history tree from **82 paths (76 branch points)** at $k = 0$ to **68 paths (66 branch points)** at every $k \neq 0$. Unreadable at bin 0 means present in full at a computable non-DC bin address.

5. Self-Clocking Transition Grammars and the Location–Shape Duality

5.1 Removing the External Clock via the $m = k + 1$ Bijection (SonicDMX)

In a transformation universe with no external reference frame, time is not a background container; time is the measure of change against a chosen recurrent transformation, and ordering is generated internally by admissible state transitions. This principle is physically instantiated and verified in the SonicDMX self-clocking acoustic protocol.

When a base-k symbol stream is subjected to the C_1 continuation constraint ($s_{t+1} \neq s_t$, forbidding adjacent identical symbols so that every symbol boundary is a self-clocking state transition), a naive base-k alphabet suffers an immediate capacity deficit: its constrained channel capacity drops to $\log_2(k-1) < \log_2(k)$, producing $21{,}324$ adjacent-equal violations in the unencoded source stream. Because Smirnov words (strings with no adjacent equal symbols) of length n over an alphabet of size m with the first symbol drawn from k values number exactly $k(m-1)^{n-1}$, achieving zero-loss fixed-length bijective encoding across all word lengths n simultaneously forces a unique alphabet size:

$$k(m-1)^{n-1} = k^n \Leftrightarrow (m-1)^{n-1} = k^{n-1} \Leftrightarrow m = k+1.$$

Exhaustive execution across $k \in \{2,3,4,5\}$ and $n \leq 8$ confirms that $m = k + 1$ is an exact bijection with zero exceptions and channel slack equal to 0.000000. To prevent an infinite regress where $k + 1$ source symbols require $k + 2$ channel symbols, the added $(k + 1)$-th symbol ($\star$, implemented as Tone 4 in SonicDMX: e.g., $111 \rightarrow 141$ and $0000 \rightarrow 0404$) carries zero independent source entropy:

Parameter / Metric	**Base k=3k = 3k=3, Channel m=3m = 3m=3**	**Base k=3k = 3k=3, Channel m=4m = 4m=4 (k+1k+1k+1)**	**Base k=3k = 3k=3, Channel m=5m = 5m=5**	**Base k=7k = 7k=7, Channel m=8m = 8m=8 (k+1k+1k+1)**
Smirnov Word Count ($n = 2,3,4,5,6$)	$6,12,24,48,96$ (Deficit)	$9,27,81,243,729$ **(Exact 3^n)**	$12,48,192,768,3072$ (Surplus)	Exact 7^n for all $n \leq 8$

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Parameter / Metric	Base k=3k = 3k=3, Channel m=3m = 3m=3	Base k=3k = 3k=3, Channel m=4m = 4m=4 (k+1k+1k+1)	Base k=3k = 3k=3, Channel m=5m = 5m=5	Base k=7k = 7k=7, Channel m=8m = 8m=8 (k+1k+1k+1)
Encoded Conditional Entropy $H(\text{enc})$	—	1.584960 bits	—	2.807336 bits
Source Entropy $\log_2(k)$ **vs** $\log_2(k+1)$	—	$\log_2(3) = 1.584963$ (vs 2.000000)	—	$\log_2(7) = 2.807355$ (vs 3.000000)
Pointer Density $P(\star)$ **vs** $\frac{1}{k+1}$	—	0.250096 vs 0.250000 (err 9.6×10^{-5})	—	0.090515 vs 0.090909 (at $k = 10$)

The fourth tone $\star$ is not a data value; it is a depth-1 history pointer (R_{past}, reference depth histogram $\{1: 50{,}152\}$) that converts temporal run duration into state-space location. Paired with the Goertzel filter bank at ≥ 19 kHz in StreamDecoderImpl, when maximum projection energy falls below LOWER_THRESHOLD, the decoder emits the non-symbol gap marker "9", latching into an idle boundary state until a valid trajectory resumes. Identical accounting governs base-b arithmetic, which runs on b digits plus 1 carry bit ($b + 1$ symbols) where $H(\text{carry} \mid x_i, y_i, \text{carry}_{\text{prev}}) = 0.0000000000$ bits. Whenever C_1 forbids a transition, the missing degree of freedom reappears as a zero-entropy state pointer occupying $\frac{1}{k+1}$ of the channel.

5.2 Location and Shape Channels in BBP, SHA-256, and the Primorial Lattice

Across the NEXUS corpus, an object is the stable residue of a constraint-satisfaction process, read through two complementary apertures: the **location channel** (the direct address in the fixed constraint field) and the **shape channel** (the boundary behavior exposed when the value is routed through an operator).

In c1_shape_channel.py, Bailey–Borwein–Plouffe (BBP) digit extraction and SHA-256 round execution are demonstrated to be the same operation—input → location in a fixed constraint field → forced readout—and both are $1{:}\,1$ in change. While SHA-256 as a full message digest is many-to-one solely because its final codomain is truncated to 2^{256}, a single SHA-256 round transformation $(a, b, c, d, e, f, g, h) \mapsto ((T_1 + T_2), a, b, c, (d + T_1), e, f, g) \pmod{2^{32}}$ is an exact bijection on $\mathbb{Z}_{2^{32}}^8$. Executing sha_round followed by sha_round_inv on random 256-bit states with schedule words K, W recovers the prior state with 100%

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bitwise identity (rec == rng). Routing a single fixed 32-bit word (x = 0x9E3779B9) through nine 32-bit SHA/bitwise read-heads (identity, ROTR2, ROTR13, Sigma0, Sigma1, Ch, Maj, XOR, AND) produces $9/9$ distinct 32-bit readouts, proving that shape is not stored inside a static noun but is exposed at the operator boundary. Furthermore, testing the hypotenuse between C_1 and the shape channel over $N = 512$ steps using both an exact irrational rotation $\theta_k = \mathrm{frac}(k\alpha)$ ($\alpha = \pi - 3$ at 60-digit precision) and the π-residue Bernoulli walk $x_{k+1} = \mathrm{frac}(16x_k)$ (at 656-digit precision) confirms that both walks remain strictly bounded in $[0,1)$ with zero linear trend ($R^2 < 0.05$). Because C_1 forbids a state-between-state straight walk across the seam, the realized traversal is a wave (the arc), while the line survives only as the direct BBP address jump (the chord).

This same frame structure is proved algebraically and verified empirically in the Primorial Family Lattice (A-Mark10). For a prime pair $(p, p+k)$ at primorial wheel depth $W = p_1 p_2 \cdots p_m$, belonging to subtype τ defined by admissible residue $p \equiv r \pmod W$, the midpoint $H = p + k/2$ satisfies the shape-channel constraint $H \equiv r + k/2 \pmod W$. By the **Step Theorem** (UNKNOWN-4 closed by induction via the Chinese Remainder Theorem), consecutive midpoints within any fixed subtype satisfy:

$$\Delta H = H_{j+1} - H_j = (p_{j+1} + k/2) - (p_j + k/2) = p_{j+1} - p_j \equiv r - r = 0 \pmod W.$$

Empirical verification across $300{,}000$ primes ($80{,}340$ total ΔH pairs across $W = 6{,}30{,}210$ and $k = 2{,}6{,}12$) confirms the Step Theorem and the Gamma-Primorial Mixture (GPM) super-Poisson spacing regularity with zero violations:

Primorial Wheel Depth WWW	**Admissible Subtypes |SW(2)|\lvert S_W(2) \rvert|SW(2)| (k=2k=2k=2)**	**Tested ΔH\Delta HΔH Pairs (k=2k=2k=2)**	**Total ΔH\Delta HΔH Pairs (k=2,6,12k=2,6,12k=2,6,12)**	**Observed Violations (ΔH≢0mod W\Delta H \not\equiv 0 \bmod WΔH≡0modW)**	**Fitted Gamma Shape Parameter αW\alpha_WαW**	**Verification Status**
$W = 6$	1	2,991	—	0	$\alpha_6 \approx 1.33 > 1$	PASS (100%)
$W = 30$	3 (6 at $k = 6$)	2,986	—	0	$\alpha_{30} \approx 1.34$ (1.38 at $k = 6$)	PASS (100%)

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Primorial Wheel Depth WWW	Admissible Subtypes |SW(2)|\lvert S_W(2) \rvert|SW(2)| (k=2k=2k=2)	Tested ΔH\Delta HΔH Pairs (k=2k=2k=2)	Total ΔH\Delta HΔH Pairs (k=2,6,12k=2,6,12k=2,6,12)	Observed Violations (ΔH≢0mod W\Delta H \not\equiv 0 \bmod WΔH≢0modW)	Fitted Gamma Shape Parameter αW\alpha_WαW	Verification Status
$W = 210$	15	2,964	80,340 (cumulative)	0	$\alpha_{210} \approx 1.37 > 1$	PASS (100%)

For any gap $k = 6d \equiv 0 \pmod 6$ at $W = 6$, the two subtypes ToA ($p \equiv 5 \pmod 6 \Longrightarrow H \equiv 3d - 1 \in \{2,5\}$) and ToB ($p \equiv 1 \pmod 6 \Longrightarrow H \equiv 3d + 1 \in \{1,4\}$) are exact runtime mirror reflections across the commit hinge $H \equiv 3d \pmod 6$, with dominant H-residues summing to $2 + 4 = 5 + 1 = 6 = W$ (and summing to $W = 30$ across all complementary subtype pairs at $W = 30$ with zero violations). While the midpoint coordinate $H \pmod W$ is frame-dependent and exhibits a Chebyshev density bias ($p = 0.005$, with frame-crossing sign flips at $k = 36{,}66$) whose infimum defines the recursion boundary $\varepsilon = H(1) \approx \pi/9 \approx 0.349066$, the gap $k = (p + k) - p$ is preserved invariantly across all subtype projections. In the SHA-256 message schedule $W[t] = \sigma_1(W[t-2]) + W[t-7] + \sigma_0(W[t-15]) + W[t-16]$, this identical ToA/ToB mirror reflection appears in the carry-amplification inversion at $W[5]/W[7]$ around the local fold $W[6]$, alongside the structural mid-fold at $W[16]$ and the 8-word Sziklai recovery window. Relativistic frame invariance ($ds^2 = -c^2dt^2 + dx^2$) is not an external physical axiom; it is the macroscopic continuum readout of the substrate's discrete subtype-projection invariance.

6. Compiled Quantum Structure, Action Extremality, and the Measurement Seam

6.1 The Division Algebra Theorem and the $C_1 a$ Independence Spine

Starting solely from the primitive constraint set $\{C_0, C_1, C_1 a, C_2, C_5\}$, the NEXUS compiler derives the necessity of complex probability amplitudes ($\mathbb{C}$), the wave equation, and Hamilton's principle ($\delta S = 0$). In *The Division Algebra Theorem* (QuHarmonics-C1-DAT-v1), $C_1 + C_1 a$ compile a linear transition state space over a scalar continuation algebra D. Because C_0 forbids nonzero states from collapsing to indistinguishability under nonzero scalar multiplication, D has no zero divisors; combined with continuous magnitude scaling over $\mathbb{R}$ and Methodological Principle M_1 (Minimal Compilation, which excludes

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unlicensed infinite-dimensional degrees of freedom), Frobenius's theorem (1877) restricts D to the three finite-dimensional real division algebras: $\mathbb{R}$ (dim1), $\mathbb{C}$ (dim2), and $\mathbb{H}$ (dim4).

The elimination of the real field $\mathbb{R}$ follows directly from C_5. Constraint C_5 requires every observable event to carry two independent degrees of freedom simultaneously: continuous spatial propagation (S) and historical residue (V). A one-dimensional real amplitude $r \in \mathbb{R}$ supports only the sign flip $r \mapsto -r$ as an intrinsic phase-like operation, generating the discrete two-element group $\mathbb{Z}_2 = \{+1, -1\}$. Because $\mathbb{Z}_2$ is disconnected, there is no continuous path between $+1$ and -1 through group elements, and it cannot realize the connected continuous phase rotation group $U(1) = \{e^{i\theta} : \theta \in [0, 2\pi)\}$ without introducing a second real degree of freedom—which is passing from $\mathbb{R}$ to $\mathbb{C}$. Thus $\mathbb{R} \notin \mathcal{A}_{C_1 a, C_5}$.

The elimination of the quaternion division algebra $\mathbb{H}$ follows from $C_1 a$ combined with the algebraic simplicity of $\mathbb{H}$. As a division algebra, $\mathbb{H}$ is a simple real algebra whose only two-sided ideals are $\{0\}$ and $\mathbb{H}$ itself. For any $\mathbb{R}$-algebra homomorphism $\rho: \mathbb{H} \to \mathrm{End}(V)$ representing scalar continuation updates, the kernel $\ker(\rho)$ is a two-sided ideal and therefore must be either $\mathbb{H}$ (the trivial zero representation, which exits $\mathbb{H}$ entirely) or $\{0\}$ (an injective, faithful representation). In every faithful representation, the image $\rho(\mathbb{H}) \cong \mathbb{H}$ inherits the non-commutativity of the quaternion basis elements i and j: applying independent scalar phase updates $T_A(q) = iq$ and $T_B(q) = jq$ to independent subsystems $A \perp B$ yields $T_A T_B(q) = i(jq) = kq$ versus $T_B T_A(q) = j(iq) = -kq$, so that the commutator $[\rho(i), \rho(j)] = 2\rho(k) \neq 0$. The joint amplitude depends on which independent subsystem was updated first, introducing unlicensed ordering structure in direct violation of $C_1 a$ ($[T_A, T_B] = 0$). Attempting to rescue $\mathbb{H}$ via a global ordering convention introduces unlicensed structure, while restricting to the commutative center $Z(\mathbb{H}) = \mathbb{R}$ exits $\mathbb{H}$. Thus $\mathbb{H} \notin \mathcal{A}_{C_1 a}$ across all representations, leaving $D = \mathbb{C}$ as the unique admissible scalar continuation algebra.

This proof exposes $C_1 a$ ($[T_A, T_B] = 0$) as the **Independence Spine** operating across three levels of the compiled architecture: at the transition operator level, it blocks hidden synchronization channels between independent regions to compile local content conservation ($C_3 \to L_3$); at the scalar amplitude level, it eliminates $\mathbb{H}$ to force $\mathbb{C}$ and its intrinsic automorphism group $U(1)$; and at the internal symmetry sector level, it requires independent gauge generators to commute ($[X_i, X_j] = 0$) so that the gauge structure factorizes into a direct product $G = G_1 \times G_2 \times \cdots$ containing $U(1)$ as its compiled scalar phase factor.

6.2 The Measurement Theorem and Action Emergence ($\delta S = 0$)

Under C_5, every continuation state decomposes into the coupled pair $s = (S, V)$, where the shape channel S carries the propagating degree of freedom ($\int \Delta V \to V$, producing the time rate $\partial_t \phi$) and the value channel V carries the historical residue ($\Delta V_{\text{spatial}} + V_{\text{history}}$, reporting spatial curvature $\nabla\phi$ back to S). By the **Measurement Theorem**, when a C_5 event commits a stable residue into the value channel, $V_{t+1} = V_t$ becomes locally stationary and enters reachable history ($H_{t+1} \supseteq H_t$ under C_2) as a **fact**, while the shape channel continues ($S_{t+1} \neq S_t$) subject to that new boundary condition, satisfying C_1 globally ($s_{t+1} \neq s_t$).

Balancing the kinetic drive of the shape channel against the spatial curvature feedback of the value channel at the compiled cost ratio c^2 yields the free-field Lagrangian density $L = (\partial_t \phi)^2 - c^2 (\nabla\phi)^2$ and the wave equation $\Box\phi = \partial_t^2 \phi - c^2 \nabla^2 \phi = 0$. In *Action Emergence* (QuHarmonics-C1-ACT-v1), each admissible discrete step $s_i \to s_{i+1}$ pays a positive local transition cost $c(s_i, s_{i+1}) = L(\phi(s_i), \partial\phi(s_i)) \Delta t \Delta V > 0$, whose sum along a path γ converges in the C_7 continuum limit to the action functional $S[\phi] = \int L \, d^4x$. Because a field

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configuration is pointwise C_1-consistent (zero uncompensated local transition residual $R(x,t) = \Box\phi(x,t) = 0$) if and only if it satisfies the Euler–Lagrange equations for L, the **Extremal Theorem** holds bidirectionally:

$$\text{Pointwise } C_1\text{-Consistent} \Leftrightarrow \Box\phi = 0 \Leftrightarrow \text{Euler–Lagrange Equations} \Leftrightarrow \delta S[\phi] = 0.$$

Hamilton's principle is not an external postulate imposed on nature; it is the global integral of local C_1-admissible binary continuation, and weighting all C_1-admissible paths by the compiled complex phase $\exp(iS/\hbar)$ yields the structural core of the Feynman path integral.

6.3 Counting the Measurement Seam: Redundancy Plateaus and the Global Wash Channel

If a quantum superposition is an unpaid transformation—a private change whose counter-transformation has not yet displaced into independent carriers under Law 3—then measurement is the first payment, and macroscopic classicality is redundant payment across multiple disjoint carriers. Because the quantum no-cloning theorem prevents multiple independent observers from reading a single copy without disturbance, consensus reality requires R disjoint environment fragments to each carry a full copy of the classical bit.

To count this ledger threshold directly, an 11-qubit exact density-matrix simulation (numpy partial-trace and von Neumann entropy computation) couples one system qubit S initially in superposition to $N_{\text{env}} = 10$ environment carrier qubits via controlled rotations parameterized by the payment depth angle θ. Evaluating environment fragments F_k of size $k \in \{1, \ldots, 10\}$ via quantum mutual information $I(S{:}F_k) = H(S) + H(F_k) - H(SF_k)$ and redundancy count $R = \lfloor 10/k_{0.9} \rfloor$ (where $k_{0.9}$ is the minimum fragment size capturing 90% of the system entropy $H(S) = 1.00$ bit) proves first that off-diagonal coherence is literally the unpaid balance of the transformation. At a shallow payment angle of $\theta = 10°$, the system retains an unpaid off-diagonal coherence balance of 0.429, whereas at full payment ($\theta = 90° = \pi/2$), off-diagonal coherence drops to 0.000 and the transformation becomes a settled classical fact. The exact same parameter that deepens the external mark extinguishes the internal superposition complementarily.

Second, the simulation confirms that the quantum-to-classical boundary is a countable redundancy threshold rather than a spatial size threshold. At full payment ($\theta = 90°$), any single carrier delivers the complete classical bit ($k_{0.9} = 1$), yielding $R = 10$ independent witnesses who can each read the history from disjoint fragments without disturbing one another. As payment depth weakens to partial ($\theta = 45°$) and weak ($\theta = 22.5° = \pi/8$) coupling, the redundancy count collapses from $R = 10 \to 3 \to 2$, spreading the mark thinner than single-fragment readability and dissolving consensus objectivity.

Third, the full-payment mutual information profile exposes the C_0 Wash channel directly inside quantum mechanics. Across every local fragment size from $k = 1$ through $k = 9$, the mutual information reads a flat classical plateau of exactly $I(S{:}F_k) = 1.00$ bit $= H(S)$; only when the fragment encompasses the entire environment ($k = 10$) does the mutual information jump antisymmetrically to $I(S{:}F_{10}) = 2.00$ bits $= 2H(S)$. That extra 1.00 bit is the quantum phase correlation, which is written across all carriers simultaneously and cancels out in every local partial trace ($k < 10$). This is the verbatim signature of C_0: maximal global correlation reading locally as zero connection. Furthermore, because Law 3 and C_4 require stepwise displacement across an adjacency graph, and position is the exact definition of graph adjacency, the coupling Hamiltonian structurally selects position-like pointer bases for redundant marking.

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7. Physical Rendering, Topological Eigenstates, and the Unified Ledger

7.1 The Rendering Program: Dual Read Heads and the Sarrus Linkage

Because the transformation field is fixed once, a physical system does not perform an exponential search over candidate trajectories; it settles into the path that satisfies its boundary constraints—a process the NEXUS program defines as **rendering** rather than calculating.

In the Orrery spatial rendering experiments (Part V of *From Nothing*), an elastic ring of nodes settles onto geometric point clouds. On the standard berlin52 benchmark ($N = 52$ cities, rounded optimum 7,542), the elastic-ring settlement arm recovers the exact optimal tour—with unrounded length $7{,}544.37$ matching the exact unrounded optimum—on $20/20$ **runs** under genuine random phase and jitter, requiring zero post-settlement 2-opt swaps. Across a multi-scale sweep, path-independence of the attractor is confirmed at all N, while the claim that the single scalar-parameter settlement arm reaches the global optimum on all geometries is falsified on st70 (687 vs 675) and structured-density pr152 (8.5% gap under $\sqrt{N}$ scaling vs 2.9% on uniform kroA100). Pairing the circular settlement read head (slack field, center-out) with its exact dual—the parameter-free linear pulled-band read head (convex hull order → cheapest interior snag → uncross, edge-in, which reduces pr152 from 8.5% to 2.4%)—bounds every tested instance within $\leq 2.5\%$ of the optimum without per-instance tuning. Furthermore, testing a sequential tension-fall (no-walker) solver where cities fall into a consumed mass center yields $1.19 \times -3.2 \times$ optimal tour length (DEAD END), proving that Law 3's closed-loop boundary condition is load-bearing: the mark carries the instruction only when the mark is the entire closed ring.

At the molecular and cryptographic scale, the **Sarrus Linkage** (1D → 3D constraint hinge) demonstrates that SHA-256 execution traces and biological protein folding share an identical geometric grammar. Extracting the 64-round T_1 execution trace of SHA-256 into 3D coordinates against a null model of $100{,}000$ random walks ($\mu = 27.77$ Å, $\sigma = 11.66$ Å) reveals three distinct topological phases bounded within the polymer compactness ratio $r_{rw} = R_g/(b\sqrt{N}) \in [0.31, 0.63]$ (matching the biological protein range 0.36 ± 0.08): Empty forms a rigid rod ($r_{rw} = 0.631, Z = +2.24$), Bitcoin ($r_{rw} = 0.310$) and Satoshi ($r_{rw} = 0.318$) form melted scrap, and GlassKey locks into a resonant knot eigenstate ($r_{rw} = 0.408, Z = -1.92, p = 0.0068$, 99.8th percentile topological closure). In biological sequences, the Sarrus autocorrelation differential $S = Z_H - Z_S$ (measuring local α-helix lags 3,4 against long-range β-sheet lag 2, mapped through Lorentz latency $L = 1/\sqrt{1 - S^2}$) predicts experimental two-state protein folding rates $\ln(k_f)$ with $r = 0.54$ **(**$p = 0.002$ **across** $n = 30$ **proteins;** L **model** AIC $= 61.4$**)** and $r = 0.6729$ **(**$p = 0.0165$ **across** $n = 12$ **Ivankov two-state folders)**.

7.2 Consolidated Program Ledger

To preserve exact dependency provenance across all domains without conflating algebraic theorems, finite-domain executions, refuted routes, and open problems, the complete record is consolidated below:

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Domain / Investigation	Specific Technical Proposition	Measured Return / Witness Certificate	Ledger Status
Foundational Cascade	C_0 $\equiv C_1, C_1a, C_2, C_3$,Law 3, C_4, C_5,Law 4	Radial opening of degrees of freedom; $\chi(M) = 0$; two-state floor $\{0,1\}$.	STANDS
Minimal Binary Substrate	Unstable vacuum C_5 derived from C_1 + propagation	$34/34$ surviving 3-cell rules satisfy $f(000) = 1$ and $f(111) = 0$.	STANDS / RUNS
Minimal Binary Substrate	Conservation forced from scalar to flux/gradient form	Only Rules 204 and 51 conserve scalar flip-count; $0/34$ overlap with survivors.	STANDS / RUNS
Minimal Binary Substrate	Second-order XOR mark layer (next $= F$(pres) $\oplus$ past)	Exact bijection on $65{,}536$ states; H(next | pres,mark) $= 0$ bits; 0% erasure.	STANDS / RUNS
Minimal Binary Substrate	Constraints select for Lempel–Ziv compression complexity	Survivor mean $0.160 <$ killed mean 0.186 across all 256 rules.	DEAD END
Native Binary Rings (R_n)	Partner involution P_n, equatorial seam, and $3/8$ boundary	$R_n[N_n - 1 - a] = P_n(R_n[a])$; $C_0(64) = 3 \cdot 2^{61} + 2^{30}$; $j_1 = 49$.	STANDS / RUNS
Native Binary Rings (R_n)	Universal bound $w^*(n) \leq 4n$ for all widths n	Width $n = 6$ alias at slices 84 and 314 shares 24 bits; $w^*(6) = 25 = 4n + 1$.	DEAD END (Bound)

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Domain / Investigation	Specific Technical Proposition	Measured Return / Witness Certificate	Ledger Status
Native Binary Rings (R_{64})	Known-phase interior vs. cross-word seam aperture (Q_{14})	Within-word requires 62 bits; seam-crossing requires 14 bits; retains 7-bit j.	RUNS
Native Binary Rings (R_{64})	Unknown-phase G_{64} synchronization ($8{,}192$ origins)	Sparse ($s = 2^{63}$): 55 samples; Contiguous grid: 127 bits; Full ring: 256 bits; retains 13 bits.	RUNS
Native Binary Rings (R_n)	Analytical proof that $w^*(n) = 4n$ for all $n \geq 8$ and d(full,full)	Verified exhaustively for $n = 8 \ldots 17$ and on G_{64} vs R_{64}, but general proof pending.	OPEN
Relation-Vocabulary Ledger	Bilinear pairing $\langle m(V), f_H \rangle$ and 3 causes of zero	9×6 multiplicity matrix on 12v witness; $m(A) = (15,4,14,4,24,36) > 0$; $14/14$ iso cap.	STANDS / RUNS
Relation-Vocabulary Ledger	Equivariant Nyquist bin recovery ($\langle z \rangle, k = 1$ bank shot)	$P_0(A) = 115, P_1(A) = 120$; spec4 and rot4 close at $k = 1$; paths drop $82 \to 68$.	STANDS / RUNS
Self-Clocking Grammar	SonicDMX C_1 Smirnov bijection and zero-entropy pointer $\star$	$m = k + 1$ exact bijection; $H(\star \mid \text{ctx}) = 0$ bits; density $\frac{1}{k+1} = 0.250096$ ($k = 3$).	STANDS / RUNS
Primorial Frame Lattice	Step Theorem ($\Delta H \equiv 0 \pmod W$) and ToA/ToB hinge mirror	CRT proof for all primorials (UNKNOWN-4); 0 violations across	STANDS / RUNS

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Domain / Investigation	Specific Technical Proposition	Measured Return / Witness Certificate	Ledger Status
		80,340 pairs (300,000 primes).	
Division Algebra & Action	$\mathbb{C}$ uniqueness from $C_1 a + C_5$; $\delta S = 0 \Leftrightarrow \Box\phi = 0$	$\mathbb{R}$ killed by C_5, $\mathbb{H}$ killed by $C_1 a$ + simplicity; Extremal Theorem compiled (cond. on C_7).	STANDS (Cond. C_7)
Quantum Measurement Seam	Superposition as unpaid balance; classicality as redundancy R	11-qubit run: R drops $10 \to 3 \to 2$; coherence $0.429 \to 0.000$; global Wash jump $1.00 \to 2.00$ bits.	RUNS
Orrery & Sarrus Rendering	Elastic ring + pulled-band dual heads; Sarrus protein/SHA trace	berlin52 7544.37 (20/20); dual heads $\leq 2.5\%$; tension-fall DEAD; Sarrus $r = 0.54, p = 0.002$.	RUNS
Frontier Strike Targets	Flux-form Law 3 filter on 34 rules; C_7 uniqueness; 3% plateau	Named next computations in the minimal substrate, geometry compilation, and Orrery linkage.	OPEN

8. Conclusion

Across every domain examined in this synthesis, a single structural architecture operates without exception: reality is constituted by binary transformations and their constrained continuations across a fixed relational space. In the minimal 256-rule binary substrate, the mandate that all states must transition (C_1) combined with bidirectional propagation compiles vacuum symmetry breaking ($f(000) = 1, f(111) = 0$ across all 34 survivors), eliminates exact single-valued scalar conservation in favor of local flux gradients (Law 3), and proves that a second-order XOR mark layer converts a first-order dream state (93.9% transient erasure) into a lossless 65,536-state reversible bijection where history carries N bits of executable instruction per step.

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When a finite reader observes this fixed binary substrate, its behavior is governed by exact quotient and realization laws. On the partner-ordered native binary ring R_{64} (2^{64} words, 2^{70} bit slices), polynomial-time canonical ranking and unranking allow an unaligned reader to evaluate at most $n = 64$ phase boundaries rather than 2^{64} word states. Crossing a word boundary invokes the native successor T_{64} and collapses known-phase contiguous acquisition from 62 bits to 14 bits, while unknown-phase synchronization across the 8,192-origin grid requires 55 sparse samples (governed by the exact $3/8$ representative boundary $C_0(64) = 3 \cdot 2^{61} + 2^{30}$), 127 contiguous bits against grid competitors, and 256 contiguous bits against the full 2^{70}-slice ring (proved sharp by a 255-bit off-grid alias), before compressing in every case to a 7-bit or 13-bit continuation-equivalence memory coordinate. On the 12-vertex relation-vocabulary ledger, every reader measurement factors bilinearly as $\langle m(V), f_H \rangle$ over the six rational irreducibles of $C_2 \times A_4$, proving that apparent zeros on the witness arise from vocabulary hyperplanes ($\phi = m(2^-) - m(1^-) = 0$) or reader symmetry blindness (V_4 on $3^{\pm}$), and that flipping the $\langle z \rangle$ reader weight from the $k = 0$ DC bin (capacity 115) to the $k = 1$ Nyquist bin (capacity 120) recovers all 120 unread dimensions, turns non-closing families (spec4,rot4) into minimal closers, and compresses the declaration history tree from 82 paths to 68.

Finally, the same binary continuation grammar removes external clocks in self-clocking channels via the $m = k + 1$ Smirnov bijection (where the added symbol is a zero-entropy depth-1 history pointer occupying $\frac{1}{k+1}$ of the channel), preserves the prime gap k as a discrete frame invariant across ToA/ToB mirror reflections and $\Delta H \equiv 0 \pmod{W}$ geodesics (verified with zero violations across $80{,}340$ prime pairs), compiles $\mathbb{C}$ and Hamilton's principle ($\delta S = 0 \Leftrightarrow \Box\phi = 0$) through the $C_1 a$ commutator spine $[T_A, T_B] = 0$, and locates the quantum-to-classical boundary at the redundancy threshold R ($10 \to 3 \to 2$) while isolating the global quantum phase surplus in the $2H(S)$ whole-environment Wash channel. Observation acquires a coordinate; the operator exposes the shape; and memory retains only the equivalence class required to continue.

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